1. Assume a and b are constants. Differentiate $4ax^2 - 3bx^{-2} + a^2$ with respect to x.

2. Show that the function
$$y = \frac{2}{1 - x^2}$$
is convex for $-1 < x < 1$. Find its global minimum and sketch the graph.

3. Find the indefinite integral:
$$\int (2x^3 + 3x - 1)dx$$

4. Let $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}$. Calculate $\mathbf{A}^2$ and $\mathbf{A}^3$

5. You would like to go downtown. On average 6 buses come by your stop every hour, and they arrive independently according to a poisson process. You arrive at the bus stop at 7:00. When you arrive, you see that you just missed three buses, they all left less than a minute earlier. How long do you think you will have to wait?

6. Suppose the PDF of a random variable X is $f_X(x) = \begin{cases} cx^{-2} \text{ if } 1 \leq x \leq 2 \\ 0 \text{ otherwise} \end{cases}$

   - What is c?

   - What is the expectation of X, $E(X)$?

7. Suppose the CIA estimates that it is able to detect a Russians cyber incursion about 90 percent of the time. That is, if the Russians hack, 9 times out of 10, the CIA will catch them. However, 20 percent of the time, when the CIA thinks it was the Russians, it was in fact some overweight guy in New Jersey. Using Bayes' rule, what is the highest prior belief about the probability of Russian hacking consistent with believing that the Russians have a less than 50 percent chance of being responsible for a hack the CIA attributes to the Russians?

8. The king is planning to go to war, but to do so, he needs to know how many soldiers he can feed. He can send auditors out into the fields to calculate the yield for plots of land, creating n independent samples $X_1 ... X_n$ chosen uniformly from across the realm. His plan is to calculate the sample mean. His advisors say that plots vary in their yield, and understand from experience that the standard deviation of plot yield is about 2 bushels.

   How large should n be so as to ensure the sample mean has a standard deviation of a week's sustenance, 0.07 bushels?